Double Bracket Flow

A Dynamical System That Diagonalizes Matrices

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Contents

1. Introduction
   - Motivation
   - Problem Formulation
2. Double Bracket Flow
   - Definition
   - Four theorems
3. Geometry
   - Isospectral Manifold
   - The Tangent Space
   - Constructing a Gradient Flow

Contents

1. Introduction
   - Motivation
   - Problem Formulation
2. Double Bracket Flow
3. Geometry

Introduction

This is a project based on R. W. Brockett's work *Dynamical systems that sort lists, diagonalize matrices, and solve linear programming problems* [1] in 1991 and *Least squares matching problems* [2]. Prof. Marek Gluza in SPMS has extended his work in quantum computing in *Double-bracket quantum algorithm for diagonalization* [3].

Motivation

Consider the problem of diagonalizing a real symmetric matrix using a continuous dynamical system.

Problem Formulation

Problem

Diagonalize the following real symmetric matrix:

$$Q = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

We seek a dynamical system whose solution converges to a diagonal matrix.

Contents

1. Introduction

2. Double Bracket Flow
   - Definition
   - Four theorems

3. Geometry

Double Bracket Flow

Definition (Doule bracket flow)

Consider the dynamical system

$$\frac{dH}{dt} = [H, [H, N]], \tag{1}$$

where H and N are real symmetric matrices, with N fixed.

Equilibrium

Theorem 1 (Equilibrium points on $SO(n)$)

Suppose that $Q = \Psi^T D_Q \Psi$ and $N = \Xi^T D_N \Xi$ with Ψ and Ξ orthogonal, $D_Q = \mathsf{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, and $D_N = \mathsf{diag}(\mu_1, \mu_2, \ldots, \mu_n)$. Assume that $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ and $\mu_1 > \mu_2 > \cdots > \mu_n$. Then the values of Θ such that $\Theta N \Theta^T Q \Theta - Q \Theta N$ vanish are the values of Θ such that

$$\widetilde{\Theta} = \Psi \Theta \Xi^T \tag{2}$$

is the product ΠD of a permutation matrix Π and a diagonal square root of the identity, D. That is, $D = \mathsf{diag}(\pm 1, \pm 1, \ldots, \pm 1)$.

Equilibrium

This theorem characterizes all equilibrium points of the gradient flow on the orthogonal group. It shows that, when both Q and N have distinct eigenvalues, the system **reaches an equilibrium** precisely when their **eigenvector directions are aligned** up to a permutation and sign changes.

In other words, the equilibrium configurations correspond to different ways of **matching the eigenvalues** of Q with those of N.

Equilibrium

Example

Consider the diagonalised matrix

$$Q = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \tag{3}$$

and a matrix from $SO(2)$

$$\Theta(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{4}$$

Equilibrium

Example (cont'd)

It follows that

$$\dot{\Theta} = -\Theta N \Theta^T Q \Theta + Q \Theta N = 2 \cos\theta \sin\theta \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} \quad (5)$$

Therefore, the equilibrium is equivalent to having

$$\theta = 0,\ \frac{1}{2}\pi,\ \pi,\ \frac{3}{2}\pi. \quad (6)$$

Equilibrium

Example (cont'd)

Hence, the four equilibria are

$$\Theta_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \Theta_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Theta_3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \Theta_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{7}$$

Note that in $SO(2)$, we have permutation matrices

$$\Pi_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \Pi_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{8}$$

Equilibrium

Example (cont'd)

It suffices

$$\Theta_1 = \Pi_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Theta_3 = \Pi_1 \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix},$$
$$\Theta_2 = \Pi_2 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \Theta_4 = \Pi_2 \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{9}$$

Convergence

Theorem 2 (Convergence to a diagonal matrix)

Suppose that N is a real diagonal matrix with unrepeated eigenvalues. If $H(0)$ is symmetric and $\dot{H} = [H, [H, N]]$, then

$$\lim_{t \to \infty} H(t) = H(\infty) \tag{10}$$

exists and is a diagonal matrix.

Convergence

It establishes the **convergence** of the double bracket flow. Starting from any symmetric matrix $H(0)$, the flow gradually eliminates its off-diagonal entries and eventually converges to a diagonal matrix, provided that N is **diagonal with distinct eigenvalues**.

Since the flow **preserves the eigenvalues** of H, the final diagonal entries must be the eigenvalues of the initial matrix $H(0)$, arranged in some order.

Thus, the double bracket flow can be viewed as a continuous method for diagonalizing symmetric matrices. However, this theorem alone does not specify the **final ordering** of the eigenvalues.

Convergence

Example

Now we go back to the initial matrix

$$H(0) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad N = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \tag{11}$$

Since $H(t)$ evolve in the space of symmetric matrices, it can be rewritten as

$$H(t) = \begin{pmatrix} a(t) & b(t) \\ b(t) & d(t) \end{pmatrix}. \tag{12}$$

Convergence

Example (cont'd)

It gives

$$[H, N] = \begin{pmatrix} 0 & -b \\ b & 0 \end{pmatrix}. \tag{13}$$

Hence,

$$[H, [H, N]] = \begin{pmatrix} 2b^2 & b(d-a) \\ b(d-a) & -2b^2 \end{pmatrix}. \tag{14}$$

Convergence

Example (cont'd)

Therefore, the dynamics of the system is equivalent to the following differential equation

$$\dot{a} = 2b^2, \qquad \dot{b} = b(d-a), \qquad \dot{d} = -2b^2. \tag{15}$$

with the initial condition being

$$a(0) = 2, \qquad b(0) = 1, \qquad d(0) = 2. \tag{16}$$

Convergence

Example (cont'd)

This is a rather complicated differential equation, the detailed process is given in the appendix. The solution for it is

$$a(t) = 2 + \tanh 2t, \quad b(t) = \operatorname{sech} 2t, \quad d(t) = 2 - \tanh 2t. \tag{17}$$

Therefore, the evolution of H is

$$H(t) = \begin{pmatrix} 2 + \tanh 2t & \operatorname{sech} 2t \\ \operatorname{sech} 2t & 2 - \tanh 2t \end{pmatrix}. \tag{18}$$

Convergence

Exmaple (cont'd)

It follows that

$$\lim_{t\to\infty} H(t) = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \tag{19}$$

which is the diagonalisation of Q_0.

Selection of Limit

Theorem 3 (Condition on convergence)

Suppose that Q and N are diagonal and have ***unrepeated eigenvalues****. With the* ***exception of certain initial points*** *contained in a finite union of codimension-one submanifolds of* $\mathrm{SO}(n)$*, all solutions of Equation (1) approach a matrix of the form ΠD with D being a diagonal square root of I and Π being that permutation matrix which reorders the diagonal of Q in such a way as to* ***maximize***

$$\eta = \sum_{i=1}^{n} q_{\pi(i)\pi(i)} n_{ii}. \tag{20}$$

Selection of Limit

It explains which equilibrium is typically selected by the gradient flow. Although there are multiple equilibrium configurations, **almost all** initial conditions lead to the configuration that **maximizes the objective function**

$$\eta = \mathrm{tr}(Q\Theta N\Theta^T).$$

By the rearrangement principle, this maximum is achieved when the eigenvalues of Q and N are **matched in the same order**, from largest to smallest.

Selection of Limit

Therefore, the flow does not merely approach an arbitrary equilibrium; for almost all initial conditions, it approaches an **optimal arrangement**. Exceptional initial conditions may converge to other equilibria.

Selection of Limit

Example

Theorem 2 guarantees the diagonalisation of the matrix, but not the sorting of eigenvalues. In the previous example, it could converge to both of the following

$$H_1 = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}. \tag{21}$$

Theorem 3 concludes that the order of the eigenvalues of N determines that of H.

Selection of Limit

Example (cont'd)

Take $Q = \text{diag}(3, 1)$, $N = \text{diag}(2, 1)$, consider the two permutation of 2 by 2 matrices, the identity **1** and the permutation $(1, 2)$,

$$\pi_1(1) = 1, \quad \pi_1(2) = 2, \quad \pi_2(1) = 2, \quad \pi_2(2) = 1. \tag{22}$$

For identity, it follows that

$$\eta_1 = q_{\pi_1(1)\pi_1(1)} n_{11} + q_{\pi_1(2)\pi_1(2)} n_{22} = q_{11} n_{11} + q_{22} n_{22} = 7, \tag{23}$$

whereas for the permutation $(1, 2)$,

$$\eta_2 = q_{\pi_2(1)\pi_2(1)} n_{11} + q_{\pi_2(2)\pi_2(2)} n_{22} = q_{22} n_{11} + q_{11} n_{22} = 5. \tag{24}$$

Selection of Limit

Example (cont'd)

It concludes that

$$\eta_1 > \eta_2, \tag{25}$$

in other words, the identity permutation maximizes the objective function. In general, for cases not in the exceptional set, the system will choose similarly ordered arrangement, i.e. the order of the diagonal elements i the objective will follow that of the matrix N chosen. As a result, in the previous example, it is $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ instead of $\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$ that is achieved.

Linear Stability

Theorem 4 (Equilibria and linear stability)

If N is symmetric and n by n with distinct eigenvalues and if $H(0)$ is symmetric with distinct eigenvalues, then $\dot{H} = [H,[H,N]]$ has $n!$ equilibrium points. The eigenvalues of the linearization of $[H,[H,N]]$ at an equilibrium point take the form

$$\big(\lambda_{\pi(i)} - \lambda_{\pi(j)}\big)\big(\mu_j - \mu_i\big), \qquad 1 \le i \le j \le n, \tag{26}$$

for some permutation π. In particular, if N is diagonal, all the eigenvalues of the linearization of $[H,[H,N]]$ at a point H_0 where H is diagonal have negative real parts if and only if the eigenvalues of H_0 and N are similarly ordered.

Linear Stability

It describes the **local stability** of the equilibrium points of the double bracket flow. When both $H(0)$ and N have distinct eigenvalues, there are exactly $n!$ equilibrium points, corresponding to all possible permutations of the eigenvalues of $H(0)$.

By linearizing the dynamical system near each equilibrium, we can determine whether small perturbations decay or grow.

Linear Stability

The theorem shows that only the equilibrium where the eigenvalues of H and N are **similarly ordered** is asymptotically stable. At this equilibrium, small perturbations decay and the system returns to equilibrium.

All other equilibria are unstable: at least one perturbation direction causes the system to move away from the equilibrium.

Hence, the double bracket flow naturally selects the correctly ordered diagonal matrix as its stable equilibrium.

Linear Stability

Example

In the previous example, the eigenvalue for $H_0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ is

$$\lambda_1 = 3, \quad \lambda_2 = 1, \tag{27}$$

where as the eigenvalues of $N = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ is

$$\mu_1 = 2, \quad \mu_2 = 1. \tag{28}$$

Linear Stability

Example (cont'd)

According to theorem 4, there should be $n! = 2$ equilibria, that is

$$H_1 = \mathsf{diag}(3, 1), \quad H_2 = \mathsf{diag}(1, 3). \tag{29}$$

For 2 by 2 matrices, there is only one off-diagonal perturbation. By choosing $i = 1$, $j = 2$, the linearization coefficient is

$$\rho_\pi = \left(\lambda_{\pi(1)} - \lambda_{\pi(2)}\right)\left(\mu_2 - \mu_1\right). \tag{30}$$

Linear Stability

Example (cont'd)

Identity perturbation. Since $\pi(1) = 1$, $\pi(2) = 2$,

$$\rho_{\pi_1} = (3-1)(1-2) = -2. \tag{31}$$

Therefore, the off-diagonal perturbation near H_1 satisfies

$$\dot{h}_{12} \approx -2h_{12}. \tag{32}$$

Linear Stability

Example (cont'd)

Hence

$$h_{12}(t) \approx h_{12}(0)e^{-2t}. \tag{33}$$

The perturbation decays to 0 as time goes. Therefore, $H_1 = \text{diag}(3, 1)$ is an asymptotically stable equilibrium.

Linear Stability

Example (cont'd)

(1,2) perturbation. For $\pi(1) = 2, \qquad \pi(2) = 1.,$

$$\rho_{\pi_2} = (1-3)(1-2) = 2. \tag{34}$$

Therefore, the off-diagonal perturbation near H_2 satisfies

$$\dot{h}_{12} \approx 2h_{12}. \tag{35}$$

Linear Stability

Example (cont'd)

Hence

$$h_{12}(t) \approx h_{12}(0)e^{2t}. \tag{36}$$

As long as the initial perturbation is non-zero, it grows exponentially. Therefore, it is a non-stable equilibrium.

Linear Stability

Compare with example 1. Note that for example 1, we have four equilibria for Θ, whereas example 4 only gives two for H. This is because Θ_1 and Θ_3 corresponds to H_1 and Θ_2 and Θ_4 corresponds to H_2 in the relation $H = \Theta^T Q \Theta$.

Contents

1. Introduction

2. Double Bracket Flow

3. Geometry
   - Isospectral Manifold
   - The Tangent Space
   - Constructing a Gradient Flow

Isospectral Manifold

Objective: Diagonalize a symmetric matrix while preserving its eigenvalues.

By the spectral theorem, consider

$$H = Q^T H_0 Q, \qquad Q \in SO(n). \tag{37}$$

The **isospectral manifold** is

$$\mathcal{O}(H_0) = \{Q^T H_0 Q : Q \in SO(n)\}. \tag{38}$$

All matrices in $\mathcal{O}(H_0)$ have the same eigenvalues.

The Tangent Space

The Lie algebra of $SO(n)$ is

$$\mathfrak{so}(n) = \{\Omega : \Omega^T = -\Omega\}. \tag{39}$$

For $\Omega \in \mathfrak{so}(n)$, the exponential map generates a curve

$$Q(t) = Qe^{t\Omega}. \tag{40}$$

This curve remains in $SO(n)$.

The Tangent Space

The corresponding curve on the isospectral manifold is

$$H(t) = e^{-t\Omega} H e^{t\Omega}. \tag{41}$$

Differentiating at $t = 0$ gives

$$\dot{H}(0) = -\Omega H + H\Omega = [H, \Omega]. \tag{42}$$

Therefore,

$$\boxed{T_H \mathcal{O}(H_0) = \{[H, \Omega] : \Omega \in \mathfrak{so}(n)\}.} \tag{43}$$

Isospectral Evolution

Since the velocity must lie in the tangent space, any smooth isospectral evolution can be written as

$$\boxed{\dot{H} = [H, \Omega(H)], \qquad \Omega(H)^T = -\Omega(H).} \tag{44}$$

The remaining problem is to choose $\Omega(H)$ so that H approaches a diagonal matrix.

Constructing a Gradient Flow

Choose a fixed diagonal matrix with distinct entries:

$$N = \mathsf{diag}(n_1, \ldots, n_n). \tag{45}$$

Define the objective function

$$F(H) = \mathsf{tr}(HN). \tag{46}$$

Its critical points on the isospectral manifold satisfy $[H, N] = 0$, implying that H is diagonal.

The Riemannian Gradient

On $SO(n)$, choose the Riemannian metric

$$g_Q(Q\Omega_1, Q\Omega_2) = -\operatorname{tr}(\Omega_1\Omega_2). \tag{47}$$

For $H = Q^T H_0 Q$, the directional derivative is

$$df_Q(Q\Omega) = -\operatorname{tr}([H, N]\Omega). \tag{48}$$

By the definition of the Riemannian gradient,

$$\boxed{\operatorname{grad} f(Q) = Q[H, N]}. \tag{49}$$

The Double Bracket Flow

The gradient ascent equation on $SO(n)$ is

$$\dot{Q} = Q[H, N]. \tag{50}$$

Using $H = Q^T H_0 Q$, we obtain

$$\begin{aligned} \dot{H} &= \dot{Q}^T H_0 Q + Q^T H_0 \dot{Q} && (51)\\ &= -[H, N]H + H[H, N] && (52)\\ &= \boxed{[H, [H, N]]}. && (53) \end{aligned}$$

Moreover,

$$\frac{d}{dt} F(H(t)) = \|[H, N]\|_F^2 \geq 0. \tag{54}$$

Geometric Derivation: Summary

1. Isospectral Manifold

Orthogonal similarity preserves eigenvalues:

$$\mathcal{O}(H_0) = \{Q^T H_0 Q : Q \in SO(n)\}. \tag{55}$$

2. Tangent Space

Admissible directions are generated by the Lie algebra $\mathfrak{so}(n)$:

$$T_H\mathcal{O}(H_0) = \{[H, \Omega] : \Omega^T = -\Omega\}. \tag{56}$$

3. Gradient Ascent

Maximizing $F(H) = \text{tr}(HN)$ with the chosen Riemannian metric yields

$$\boxed{\dot{H} = [H, [H, N]]}. \tag{57}$$

Proof for theorem 1

Proof of Theorem 1. (Compare with Ref. [2]). Without loss of generality we can suppose that Q and N are diagonal. Because the ijth element of $NP - PN$ is $(n_{ii} - n_{jj})p_{ij}$, we see that $NP - PN$ vanishes only when P is diagonal. Thus $N\Theta^T Q\Theta - \Theta^T Q\Theta N$ vanishes only if $\Theta^T Q\Theta$ is diagonal. If Q is diagonal and if there is a diagonal matrix D such that $\Theta^T Q\Theta = D$, then D has the same eigenvalues as Q and $Q\Theta = \Theta D$. For this to be true we must have

$$\lambda_i \theta_{ij} = \lambda_{\pi(j)} \theta_{ij}, \qquad 1 \leq i, j \leq n, \tag{58}$$

for some permutation π of the indices $(1, 2, \ldots, n)$ whereby $i \mapsto \pi(i)$. This means that either $\lambda_i = \lambda_{\pi(j)}$, so that $i = \pi(j)$, or $\theta_{ij} = 0$. Fixing i, we see that $\pi(j)$ equals i for exactly one value of j and so only one θ_{ij} can be nonzero. Because the rows of Θ are unit vectors, the nonzero value must be ± 1. From this all else follows. ■

Proof for theorem 2

Proof of Theorem 2. First of all, because $[H, N]$ is skew-symmetric, $[H, [H, N]]$ is symmetric and this equation evolves in the space of symmetric matrices. A direct calculation shows that

$$\begin{aligned} \frac{d}{dt} \operatorname{tr} HN &= \operatorname{tr} N[H, [H, N]] \\ &= -\operatorname{tr}(HN - NH)^2. \end{aligned} \tag{59}$$

Because $HN - NH$ is skew-symmetric, this is expressible as

$$\frac{d}{dt} \operatorname{tr} HN = \operatorname{tr}(HN - NH)(HN - NH)^T. \tag{60}$$

Thus tr HN is monotone increasing and is bounded from above. Therefore, it has a limit and its derivative goes to zero. However, its time derivative vanishes only if H and N commute, and since N has distinct eigenvalues, $HN = NH$ if and only if H is diagonal. Thus H approaches a diagonal matrix. ■

Proof for theorem 3

The proof of this is given in Ref. [2].

Proof for theorem 4

Proof of Theorem 4. As in the proof of Theorem 1, we can make a preliminary transformation to put N in diagonal form. From Theorem 2 we see that H approaches a diagonal matrix which must be of the form

$$H = \operatorname{diag}\big(\lambda_{\pi(1)}, \lambda_{\pi(2)}, \ldots, \lambda_{\pi(n)}\big). \tag{61}$$

Near the diagonal we can linearize $[H, [H, N]]$ as

$$\dot{h}_{ij} = \big(\lambda_{\pi(i)} - \lambda_{\pi(j)}\big)\big(\mu_j - \mu_i\big)h_{ij} \tag{62}$$

for $i > j$. ■

Solution for the differential equation

Solution to the differential equation. We solve the system

$$\dot{a} = 2b^2, \qquad \dot{b} = b(d-a), \qquad \dot{d} = -2b^2, \tag{63}$$

subject to

$$a(0) = 2, \qquad b(0) = 1, \qquad d(0) = 2. \tag{64}$$

Solution for the differential equation (cont'd)

First, observe that

$$\frac{d}{dt}(a+d) = \dot{a} + \dot{d} = 2b^2 - 2b^2 = 0. \tag{65}$$

Hence $a + d$ is constant. Using the initial conditions, we obtain

$$a(t) + d(t) = 4. \tag{66}$$

Solution for the differential equation (cont'd)

Define

$$x(t) = \frac{a(t) - d(t)}{2}. \tag{67}$$

Then

$$\dot{x} = \frac{\dot{a} - \dot{d}}{2} = 2b^2. \tag{68}$$

Moreover, since $d - a = -2x$, the equation for b becomes

$$\dot{b} = -2xb. \tag{69}$$

Solution for the differential equation (cont'd)

We now compute

$$\frac{d}{dt}\left(x^2+b^2\right) = 2x\dot{x} + 2b\dot{b} = 4xb^2 - 4xb^2 = 0. \tag{70}$$

Therefore, $x^2 + b^2$ is conserved. Since

$$x(0) = 0, \qquad b(0) = 1, \tag{71}$$

we have

$$x^2 + b^2 = 1. \tag{72}$$

Solution for the differential equation (cont'd)

Thus,

$$b^2 = 1 - x^2, \tag{73}$$

and consequently

$$\dot{x} = 2(1 - x^2). \tag{74}$$

Separating variables gives

$$\frac{dx}{1 - x^2} = 2\,dt. \tag{75}$$

Using $x(0) = 0$, we obtain

$$x(t) = \tanh(2t). \tag{76}$$

Solution for the differential equation (cont'd)

Since $b(0) = 1 > 0$, it follows from $x^2 + b^2 = 1$ that

$$b(t) = \mathrm{sech}(2t). \tag{77}$$

Finally, using

$$a + d = 4, \qquad a - d = 2x, \tag{78}$$

we find

$$a(t) = 2 + \tanh(2t) \tag{79}$$

and

$$d(t) = 2 - \tanh(2t). \tag{80}$$

Therefore, the solution is

$$a(t) = 2 + \tanh(2t), \qquad b(t) = \mathrm{sech}(2t), \qquad d(t) = 2 - \tanh(2t). \tag{81}$$

■

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